

Comparison of Methods for Voltage Sag Source Location in Distribution System

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Abstract—Voltage sag has been one of the most concerned problems of power quality. Identifying the location of the source of voltage sag is critically necessary for power-quality troubleshooting, diagnosis and mitigation, and it is helpful to distinguish responsibilities and solve disputes caused by voltage sag. This paper analyses and compares four existing location methods based on reactive power change, slope of the system trajectory, real current component and distance relay. The distance relay method shows the best performance through simulations using PSCAD.

Keywords—power quality; voltage sag source; location

I. INTRODUCTION

Recently, highly automated equipments and power electronic devices are widely used in industry, but they are more vulnerable to the impact of voltage sag compared with traditional equipments, a few circles of voltage sag could affect these equipments work normally, and cause huge economic loss. Therefore, voltage sag has gradually become a kind of most severe dynamic power quality problem. According to foreign investigations: voltage sag of the power quality problems has become a major cause for complaint, and even accounted for 80% of the complaints. To decrease the impact of sags as much as possible, the first step is to introduce effective method for voltage sag source location detection. It is necessary for detection, diagnosis, formulating mitigating devices of power quality faults to locate the source of voltage sags, and it is helpful to distinguish responsibilities and solve disputes between power department and customs [1-3].

At present, the domestic and overseas methods on voltage sag location mainly include the methods based on disturbance power and energy, real current component, resistance sign [4], distance relay, causes of the event [5] and sag magnitude and the phase-angle jump [6]. The reactive power change method applied in this paper is an alternative method of disturbance power and energy. The four methods based on reactive power change, slope of the system trajectory, real current component and distance relay are analyzed and compared and respective characteristics of the methods are discussed. The distance relay method shows the best performance with the PSCAD simulations and the further research of the method is proposed.

II. VOLTAGE SAG SOURCE LOCATION METHODS

Voltage sag source location is to determine the fault causing the voltage sag on which side of the monitoring device [7]. As Fig.1 shown below, referring to the direction of fundamental active power, if the fault occurred at point A, in the rear of monitoring device, it is called the upstream direction; instead, if the fault occurred at point B, in front of monitoring device, it is called the downstream direction.

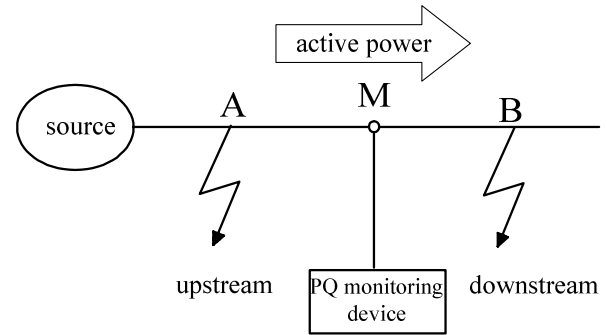


Figure 1. Schematic diagram of voltage sag source detection

A. Reactive power change method

The earlier method aiming to locate the source of disturbances is based on the disturbance power (DP) and the disturbance energy (DE) proposed in [8]. The disturbance power (DP) is the difference between the power delivered during the fault $p(t)_f$ and during the steady-state conditions $p(t)_{ss}$. The disturbance energy (DE) is the integral of the disturbance power over the disturbance event duration.

$$DP = p(t)_f - p(t)_{ss} \quad (1)$$

$$DE = \int DP(t) dt \quad (2)$$

As the disturbance occurs, part of energy tends to flow towards the disturbance source, energy and power will change. Information about changes in the DE combined with the sign of

the initial peak of the DP allows us to determine the sag source relative to monitoring points.

The approach based on the variation of reactive power used in [9] is an alternative method of disturbance power and energy. It is expected that the variation of the reactive power is an indication of the location of the sag source. If there is a positive deflection of reactive power during the disturbance and the change trend of reactive power is the same as that of active power $Q/P > 0$, Q is reactive power, P is active power, then the sag source is considered downstream, otherwise the sag source is upstream. But this kind of methods only based on the intuitive understanding of voltage sag to determine the sag source location, lacking rigorous mathematical analysis of algorithm.

B. Slope of the system trajectory method

The method of the slope of the system trajectory in [10] is based on the relationship between the voltage and the current during the fault. When the voltage sag occurs, the relationship between the product of voltage magnitude and power factor against current magnitude at the measurement location are not the same for different fault locations. Applying least-square method to fit a straight line to the data of $|U \cos \phi|$ and I during sag, the sign of this line indicates the direction of voltage sag source.

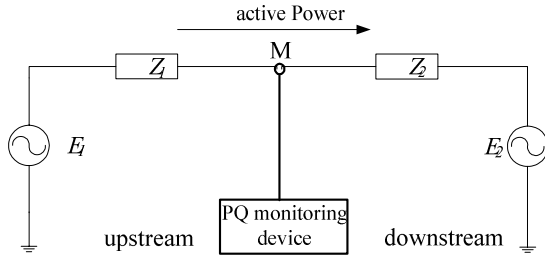


Figure 2. Equivalent circuit for sag source location analysis

As shown in Fig.2, considering that a fault occurs downstream of the monitoring point, referring to the fundamental active power flow direction, the active power measured at the PQ monitor is

$$UI \cos \phi = -R_1 I^2 + E_1 I \cos \theta_1 \quad (3)$$

Where ϕ the phase difference between U and I , θ_1 is the phase difference between E_1 and I , R_1 is the real part of Z_1 .

If $\cos \phi > 0$, $|U \cos \phi| = U \cos \phi$, $\cos \theta_1$ does not change greatly during the disturbance, the each point of $(I, |U \cos \phi|)$ is at a certain line with slope of $-R_1$.

$$|U \cos \phi| = -R_1 I + E_1 \cos \theta_1 \quad (4)$$

On the contrary, if $\cos \phi < 0$, then

$$|U \cos \phi| = +R_1 I + E_1 \cos \theta_1 \quad (5)$$

each point of $(I, |U \cos \phi|)$ is on a certain line with slope of $+R_1$.

If the line connecting the points of $(I, |U \cos \phi|)$ has a positive slope, then the fault is upstream, negative slope indicates a downstream fault. If the active power flow reverses during the fault, the voltage sag source is upstream. The method only based on the sign of the slope and there is no need to calculate other parameters or set a threshold, so it is simple and easy to implement, and it includes large data, this will enhance its credibility. But the method may not reach a conclusion for different signs of the slope for each phase.

C. Distance relay method

The method in [11] applies protective relays to power quality analysis. Most of the transmission lines are installed with distance relays, it judges the location from the information on impedance magnitude and phase angle change before and after sag, which are obtained from the distance relay or its algorithm. The impedance is estimated by the voltage and current phasors. The rule is:

If $|Z_{sag}| < |Z_{presag}|$, and $\angle Z_{sag} > 0$, then the sag source is located downstream, otherwise the sag source is located upstream.

This method has some limitations, fault location and other factors will affect the judgments, such as in a radial system. If a fault occurs between the source and distance relay, there will be no change in impedance and no conclusive result; when a fault is not permanent, the method may not derive a decision.

D. Real current component method

The method proposed in [12] points out that the relationship between the product of current magnitude and power factor at the measuring point are not the same for different fault locations. A graph of the product $I \cos \phi$ against time t is plotted to judge the sag source location.

From the Fig.2, for a fault downstream, according to equation (3), as the fault occurs, the current at monitoring point M flows from E_1 to the fault location, $I \cos \phi > 0$, the fault is said to coming from downstream relative to the point M. This current is in the same direction as the pre-fault current from E_1 to E_2 . For a fault upstream, the current flows to ground at the fault point and direction is from E_2 to the fault location, $I \cos \phi < 0$, the fault is said to occur upstream.

If at the beginning of a fault, $I \cos \phi > 0$, then the fault is from downstream. On the other hand, if $I \cos \phi < 0$, the fault is located upstream. Basically, the method investigates the power factor $\cos \phi$ at the beginning of the voltage sag, power factor and other factors will affect the judgment. In this method, the fault impedance is taken into consideration, for $Z_2 \gg Z_f$, Z_f is fault impedance, and then no current is flowing from E_2 since all current flows from the E_1 to fault impedance.

III. SIMULATIONS

Voltage sag events caused by symmetric and asymmetric faults are taken as an example for sag source location analysis

in this paper. Radial distribution system construction for voltage sag simulation shown in Fig.3 is modeled using PSCAD/EMTDC power system simulation software. Short-circuit capacity of system is 3604MVA on 220kV system substation side, main transformer capacity of 220kV substation is 150MVA. Load 1 is a nonlinear six-pulse rectifier with the capacity of 11MVA, power factor is 0.96 and the length of power lines is 30km. Load 2 is an electric locomotive, the capacity of the traction transformer station with V/V connection is $2 \times (20+16)$ MVA, power supply arm phase is AC/BC and the power line length is 34km. Load 3 is a linear load with capacity of 53MVA, power factor is 0.9 and the length of power lines is 10km. Three fault points F1, F2 and F3 are considered and three monitoring points are marked as M1, M2 and M3. The position of load 1 access PCC is set as M1, M2 is set at the point of load 1 access PCC, M3 is at the position of load 3 access PCC.

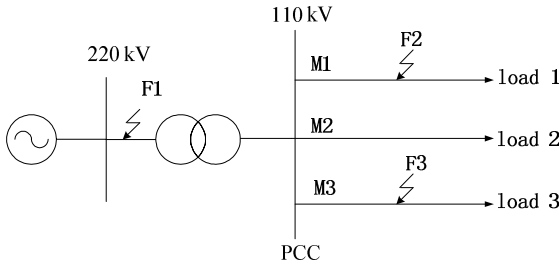


Figure 3. Radial distribution system construction for voltage sag simulation

Different fault types, symmetrical fault (three-phase short circuit) and asymmetric faults (single-phase ground, double-phase short circuit ground, double-phase short circuit) are set at fault points. The four methods based on reactive power change, slope of the system trajectory, distance relay and real current component are implemented with the data obtained from the monitoring point and the results of the four methods are discussed and compared.

A. Asymmetrical faults

A single-line-to-ground fault in phase A is set to occur at 0.25s in the fault point F3 and the duration of the fault is 0.1s. Before and after the fault, RMS voltage curve at PCC and the results of reactive power change method at monitoring points M1-M3 are presented in Fig.4 and Fig.5.

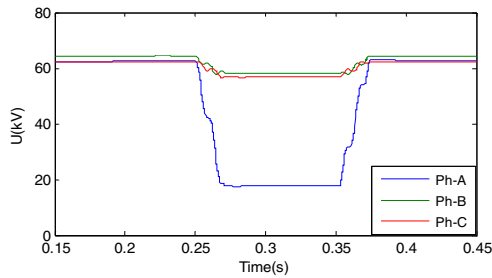


Figure 4. RMS voltage curve at PCC

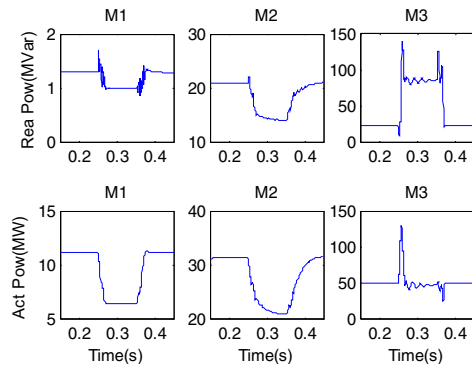


Figure 5. Results of reactive power change method at monitoring points M1-M3

In Fig.4, voltage sag occurs in phase A at PCC point during the fault, the amplitude of voltage decreases to about 30% of rating. It can be seen in Fig.5 that the reactive power at M3 has a positive change during the sag and the active power changes in the same direction, meaning the fault is located downstream. On the other hand, the reactive power at M1 and M2 decreases rapidly during the sag, the fault is upstream, which is consistent with the simulation settings.

The magnitude and angle change of impedance applying the method of distance relay at monitoring points M1-M3 is shown in Fig.6.

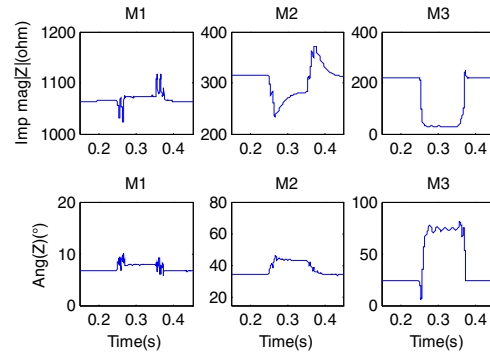


Figure 6. Results of distance relay method at monitoring points M1-M3

In Fig.6, it can be seen that the magnitude of the impedance at M3 reduces rapidly while its angle increases significantly during the sag, this signifies a downstream fault. At M1 and M2, the magnitude and angle of the impedance almost remains the same in pre- and during-sag period, this signifies an upstream fault.

The method of the slope of the system trajectory is applied and the graphs of the system trajectory in phase A, B and C at monitoring point M1-M3 are shown in Fig.7. A straight line is fit to the data of $(I, |U\cos\phi|)$ by least-square method. It is expected to obtain the same sign of the slopes of the lines in

phase A, B and C at monitoring points to judge the sag source location and it may not reach a conclusion for different signs of the slope for each phase.

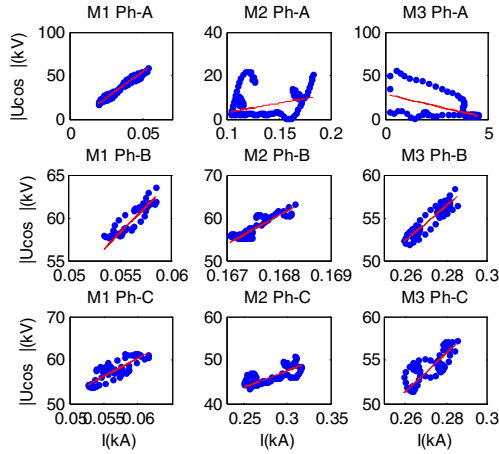


Figure 7. Results of slope of the system trajectory method at monitoring points M1-M3

From the Fig.7, the slopes of the system trajectory are positive and consistent in sign at monitoring M1 and M2, indicating the sag source is from upstream. At M3, the slope of the line in phase A is negative and the slopes in phase B and C are positive, it can not make a conclusion because there is not sign agreement on the slopes.

Fig.8 shows the graphs of real current component plotted against time during the fault at monitoring points M1-M3.

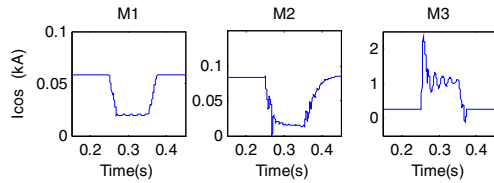


Figure 8. Results of real current component method at monitoring points M1-M3

In Fig.8, it can be seen that the real current component at M3 changes substantially at the beginning of the sag, the real current settles out to a positive value and change, the fault is classified as downstream relative to the monitoring point M3. The real current at M1 and M2 decreases a little at the beginning of the sag, the amplitude of variation is very small relative to monitoring M3, which signifies the direction of the current at M1 and M3 has not changed, the sag source is located upstream.

B. Symmetrical faults

A three-phase short circuit fault is set to occur at 0.25s in the fault point F3 and the duration of the fault is 0.1s. The magnitude and angle change of impedance applying the

method of distance relay before and during the sag at monitoring points M1-M3 is shown in Fig.9.

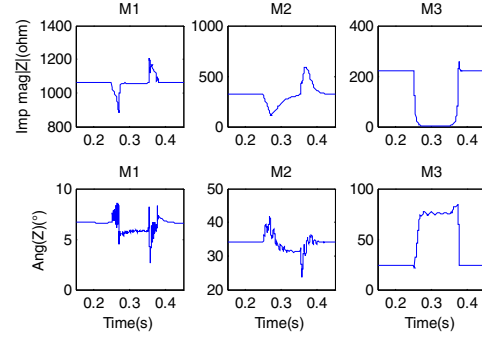


Figure 9. Results of distance relay method at monitoring points M1-M3

As shown in Fig.9, the magnitude of the impedance at M3 reduces substantially while its angle increases significantly during the sag, which indicates the sag source coming from downstream. At M1 and M2, the magnitude and angle of the impedance almost remains the same before and during the sag, indicating an upstream sag source location.

For symmetrical faults located at F3, the results applying the method of reactive power change slope of the system trajectory and real current component at M1-M3 are presented in Table I .

TABLE I. RESULTS OF THE OTHER THREE LOCATION METHODS

Location Method	M1	M2	M3
RP	Upstream	Upstream	Downstream
SST	Upstream	Upstream	Downstream
RCC	Upstream	Upstream	Downstream

RP: reactive power change method, SST: slope of the system trajectory method, RCC: real current component method

It can be seen from the locating results in Table I that the three methods above accurately judge the location of the sag source for symmetrical faults.

Changing the fault location and fault type, symmetrical faults and asymmetrical faults are created at F2 and F3, asymmetrical faults are created at F1, and then thirty-three groups of sag data can be got, including twenty-seven groups of asymmetrical faults sag data and six groups of symmetrical faults sag data. The four methods based on reactive power change, slope of the system trajectory, distance relay and real current component are applied to determine the upstream and downstream location with the data obtained from the monitoring points, and the simulation results are shown in Table II and TableIII.

TABLE II. RESULTS OF SYMMETRICAL AND ASYMMETRICAL FAULTS

Fault Type	Simulation Number	Correct Number			
		RP	DR	SST	RCC
Asymmetrical faults	27	24	27	18	20
Symmetrical faults	6	6	6	6	6

TABLE III. RESULTS OF FOUR LOCATION METHODS

Location Method	Simulation Number	Correct Number	Accuracy (%)
RP	33	30	91
DR	33	33	100
SST	33	24	73
RCC	33	26	79

RP: reactive power change method, DR: distance relay method, SST: slope of the system trajectory method, RCC: real current component method

According to the results in Table II and Table III, the performance of the four location methods is accurate for symmetrical faults while relatively less accurate behavior was got in the case of asymmetrical faults. The method of distance relay shows the best performance in determining the sag location, and the accuracy of the slope of the system trajectory method is 66.67% which is not satisfactory when judging asymmetric faults. The reactive power change method performs better but is affected by the distance between the monitoring points and fault points. Power factor and fault impedance and other factors will affect the performance of the real current component method, which may reduce the credibility of the criteria and result in a lower accuracy.

The distance relay method is based on the change of the magnitude and angle of impedance during the sag to judge the location, the magnitude and angle of impedance of the branch have substantial change for sags due to downstream fault, while the impedance magnitude and angle change smaller for sags caused due to upstream fault. For a single-line-to-ground fault located at F3 in Fig.6, the magnitude and angle change values of impedance are presented in Table IV.

TABLE IV. MAGNITUDE AND ANGLE OF IMPEDANCE AT MONITORING POINTS M1-M3

	$ Z_{presag} $ (ohm)	$ Z_{sag} $ (ohm)	$\Delta Z $ (ohm)	$\Delta\angle Z$ (°)
M1	1062.3	1073.4	11.1	1.3
M2	318.2	283.7	-34.5	8.7
M3	219.9	41.5	-178.4	45.2

$|Z_{presag}|$: impedance magnitude before the sag, $|Z_{sag}|$: impedance magnitude during the sag, $\Delta|Z|$: impedance magnitude change, $\Delta\angle Z$: impedance angle change.

It is observed from Table IV that the impedance magnitude increases at M1 and reduces at M2 during the sag, the impedance angles of the two monitoring points increase, but both of the impedance magnitude and angle change are very small relative to monitoring point F3, the fault at M1 and M2 is classified upstream. The criteria may be affected by the

amplitude of the change of the impedance magnitude and angle and it is possible to choose a more reasonable threshold through a large number of simulations and analysis of the data collected from the actual monitoring points, so as to improve applicability of the detection method and accuracy of the detection results.

IV. CONCLUSIONS

Four location methods applied to locate the voltage sag source for simulating different fault positions and types have been analyzed and compared. In general the judgments of the methods above are accurate for symmetrical faults. On the other hand, the results of the methods show a relatively lower accuracy for asymmetrical faults. The distance relay method performs best in locating the sag source, and the accuracy is the highest. But the criteria may be affected by the amplitude of the change of the impedance magnitude and angle and it is possible to choose a reasonable threshold through simulations and analysis of actual data to improve the accuracy of the detection results.

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